# FI Trading Platform

## Technical Documentation

Django-based trading operations, execution, market-data and portfolio analytics platform

| | |
|---|---|
| **Technology** | Django / Python |
| **Broker Integration** | Interactive Brokers (IBKR) |
| **Core Users** | Super Admin, Admin, Portfolio Manager |
| **Interface** | Three-column trading dashboard with light and dark modes |

**Purpose.** This document describes the implemented functionality, user workflow, market-data and execution capabilities, analytics, controls, and operational monitoring demonstrated in the accompanying project video.

# 1. Project Overview

FI Trading Platform is a Django-based trading application built around the operational needs of portfolio managers. It brings market monitoring, manual execution, order management, algorithmic controls, portfolio analytics, account information and system monitoring into a single trading workspace.

The platform integrates with Interactive Brokers (IBKR) for live account and trading information and provides real-time market views, Level 2 order-book information, live charts, P&L monitoring and execution controls.

### Primary capabilities

- Live market data and Level 2 order-book visualization.
- Manual market and limit order execution with GTC/GTD and RTH/time-related controls.
- Saved orders and open-order editing.
- Instrument-specific price alerts with dashboard notifications and SMTP email delivery.
- Day high, day low and previous US-session close reference information.
- Live positions, strategy positions and saved-order monitoring.
- Live charts across instruments and multiple timeframes.
- Trading-session volume analysis for Asia, London, US and custom windows.
- Algorithm Controller with maximum quantity protection and entry/exit slippage controls.
- Portfolio heatmaps, cumulative MTM and strategy-level performance analytics.
- IBKR account-value and P&L integration, including daily, realised/unrealised, YTD and MTD views.
- Activity, trade-log and system-error monitoring.
- Role-based access for Super Admin, Admin and Portfolio Manager.
- Light and dark interface modes.

# 2. Technology & Integration

## Application technology

- Django and Python for the web application and backend.
- Trading-oriented dashboard interface with live data presentation.
- SMTP integration for automated price-alert email notifications.
- Interactive Brokers integration for account, market, order and P&L information.

## Interactive Brokers integration

IBKR is used as the external trading and account-data integration layer. The application surfaces broker-side information inside the trading dashboard, including account value, positions, order activity and P&L information.

## SMTP integration

The price-alert workflow uses SMTP to deliver automated email notifications when configured price conditions are crossed.

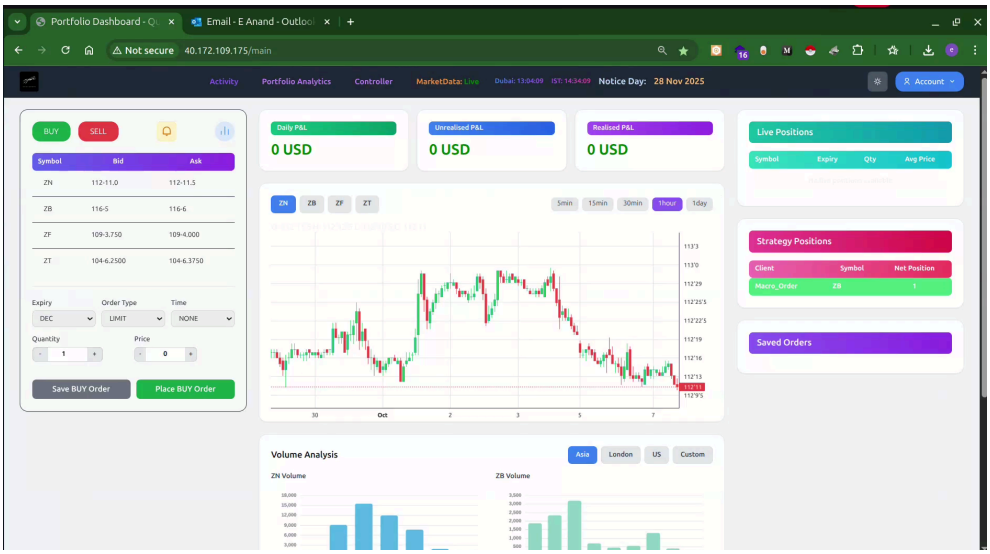
# 3. User Roles

| Role | Primary responsibility |
|---|---|
| Super Admin | System-level administration and oversight. |
| Admin | Administrative and operational management. |
| Portfolio Manager | Market monitoring, order execution, positions, alerts, analytics and algorithm controls. |

# 4. Main Trading Dashboard

The dashboard is the central workspace. Its navigation exposes Activity, Portfolio Analytics and Algo Controller, while the header provides market-data connection status, Dubai time, IST, trading-day information and notices. Account information is available through a separate account section.

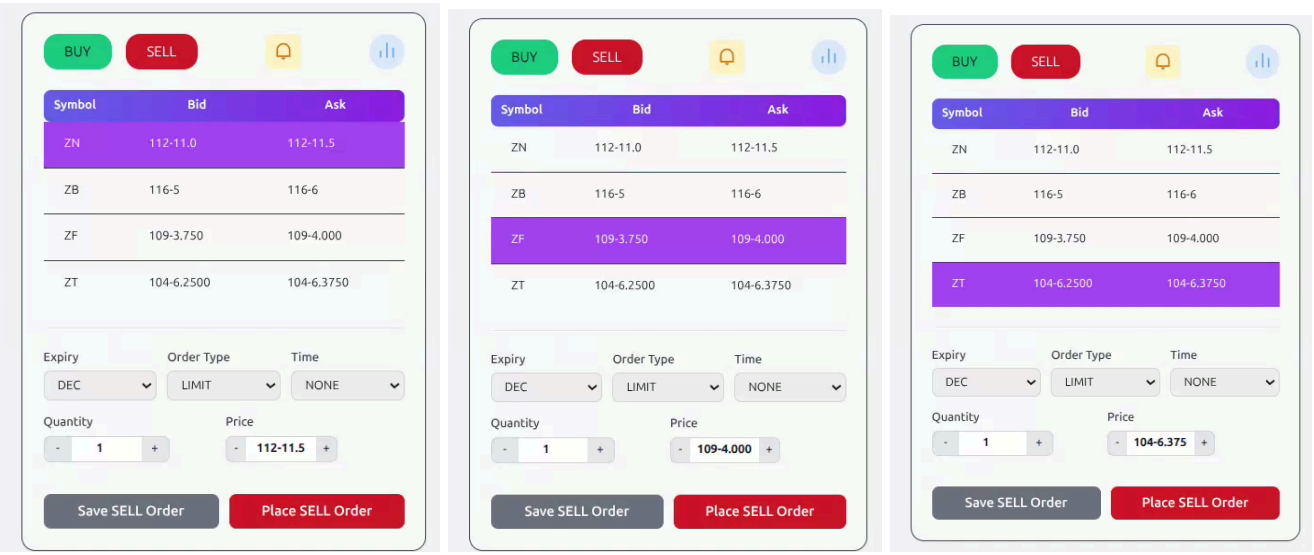
The primary dashboard follows a three-column trading layout: the first column focuses on trading and order entry; the center column combines P&L, live charting and volume analysis; the third column provides live positions, strategy positions and saved orders.



# 5. Trading & Order Management

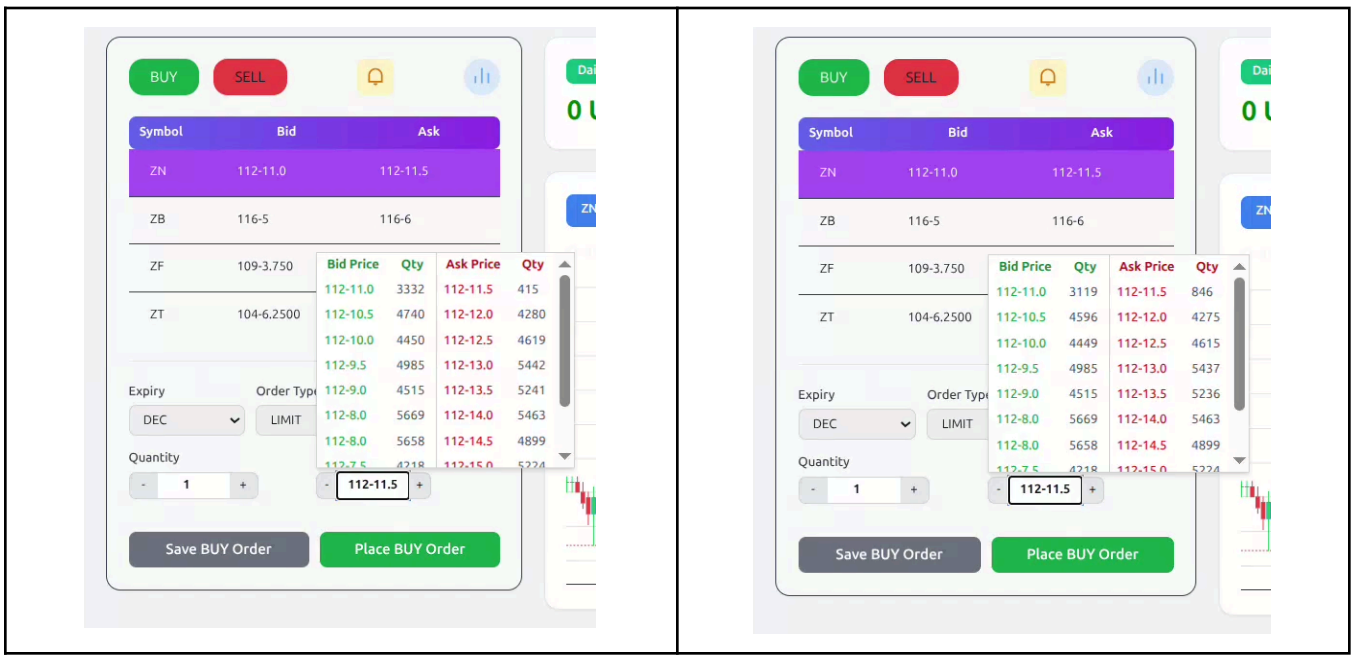
## Instrument selection and live bid/ask

- Instrument selection directly updates the trading interface.
- Live bid and ask prices are displayed with corresponding quantities.
- The buy/sell workflow uses the relevant market side automatically.
- Live Level 2 order-book information is available for both bid and ask sides.




## Level 2 order book

- Bid-side and ask-side market depth can be viewed.
- The interface supports visibility of up to 20 levels on each side.
- Live price and quantity are shown for each order-book level.
- Market-data subscriptions are controlled to remain within the application's subscription constraints.




**LEVEL 2 ORDER BOOK**

### Manual order execution

- Market orders.
- Limit orders.
- GTC (Good-Till-Cancelled) orders.
- GTD (Good-Till-Date) orders.
- RTH/time-related order configuration.
- Quantity increase/decrease controls.
- Expiry and time selection where applicable.

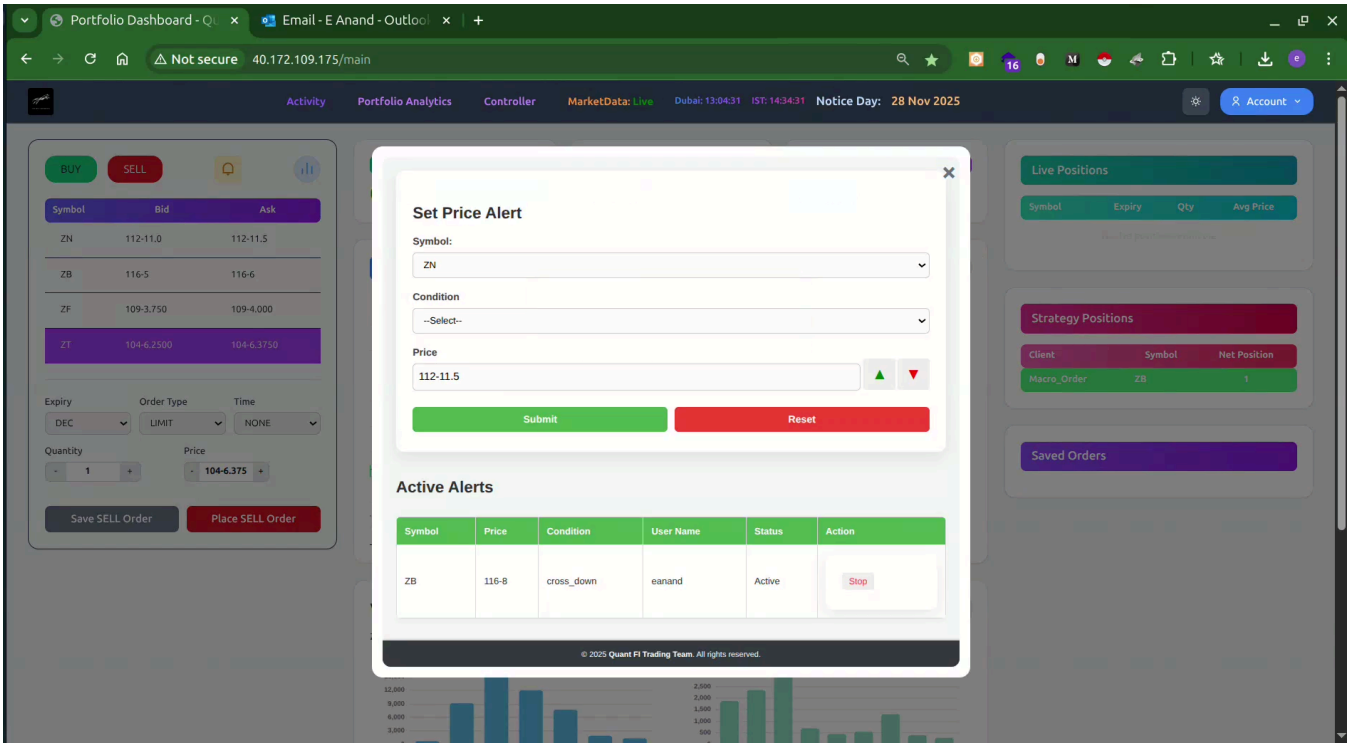
### Saved and open orders

- Orders can be prepared and saved before execution.
- Saved orders can subsequently be placed.
- Open orders can be edited from the activity/order-management workflow.

## 6. Price Alert System

The platform supports instrument-specific price alerts. A Portfolio Manager can configure a target price and direction. When the market crosses the configured level, the system records the alert, shows a dashboard notification and sends an automated email through SMTP.

Alert flow: Configure price → monitor live market price → detect upward/downward crossing → record alert → dashboard notification → SMTP email.

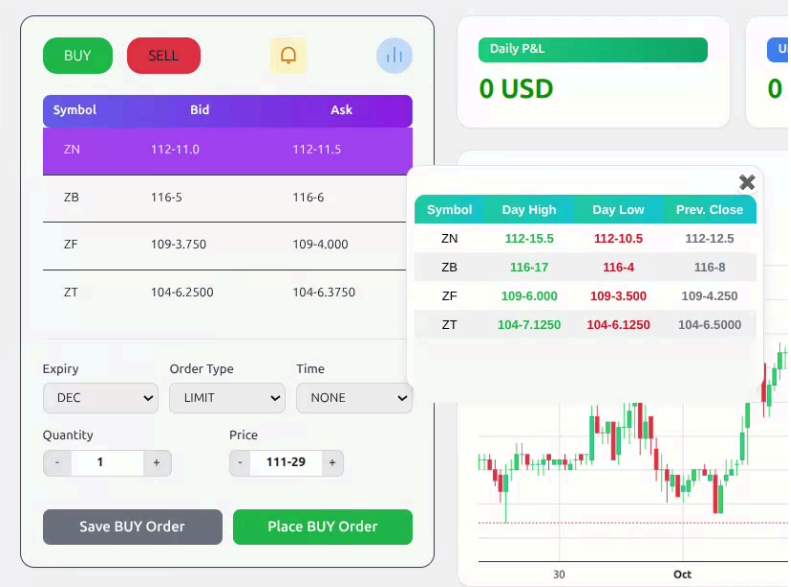



**PRICE ALERT NOTIFICATION AND MAIL SYSTEM**

# 7. Market Reference & Session Analytics

## Day high, day low and previous close

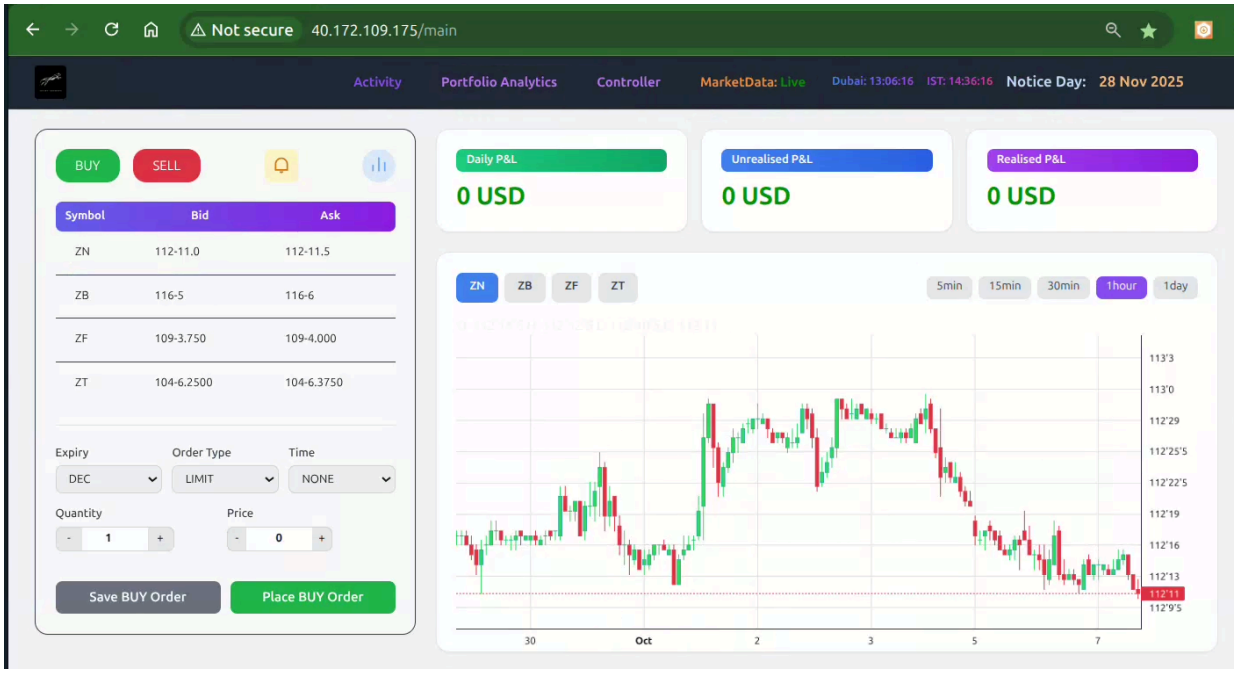
- Instrument-specific day high.
- Instrument-specific day low.
- Previous US-session close reference.




**Day high, day low and previous close**

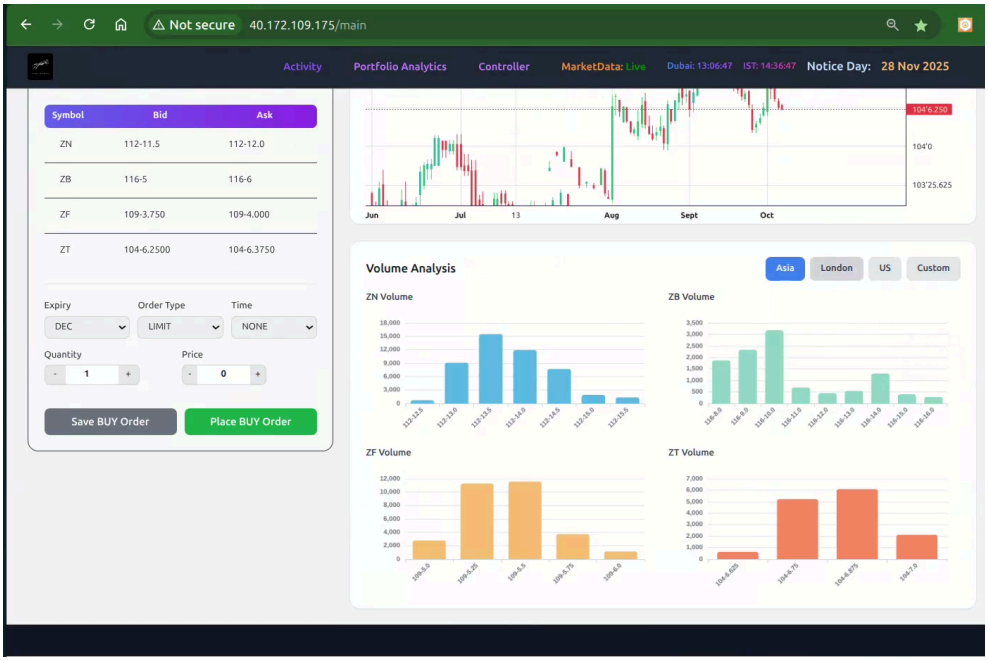
## Live charting

- Multiple instruments can be selected.
- Multiple chart timeframes are supported.
- Charts consume live tick data.
- Different timezone views are supported for trading analysis.



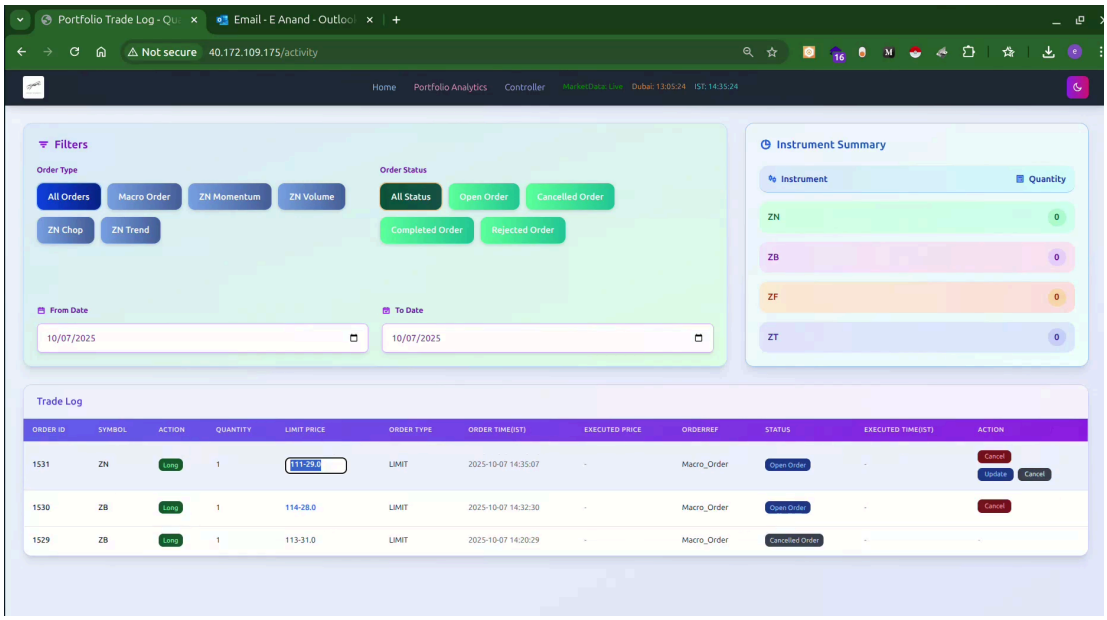
### Volume analysis

- Asia trading session.
- London trading session.
- US trading session.
- Custom user-defined analysis window.



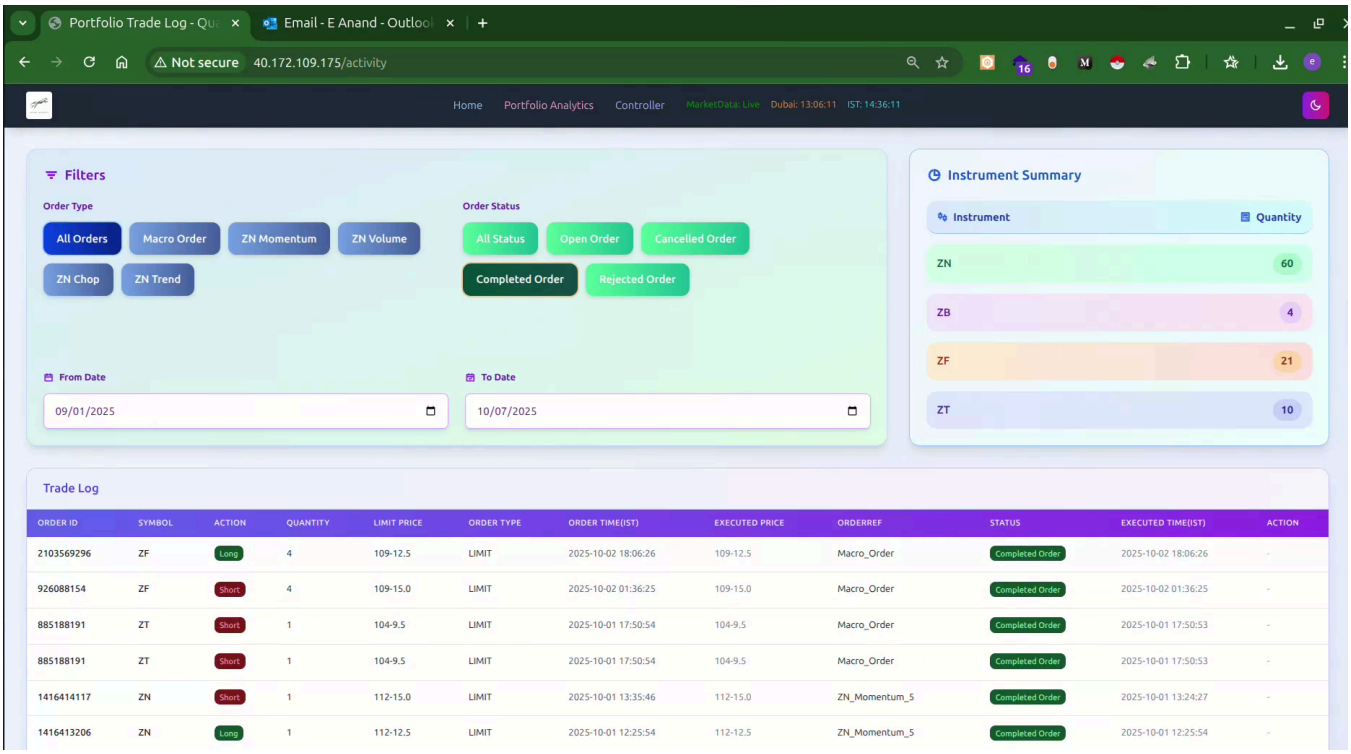

# 8. Activity & Order Monitoring

- Activity history with filtering.
- Trade log with instrument and execution summaries.
- Key-value details for operational inspection.
- Open-order views and filtering.
- Editing of eligible open orders.



# 9. Live Positions & Strategy Positions

The dashboard separates current live account positions from strategy-level positions and saved orders. This gives the Portfolio Manager simultaneous visibility into actual exposure, strategy activity and pending or prepared trading actions.



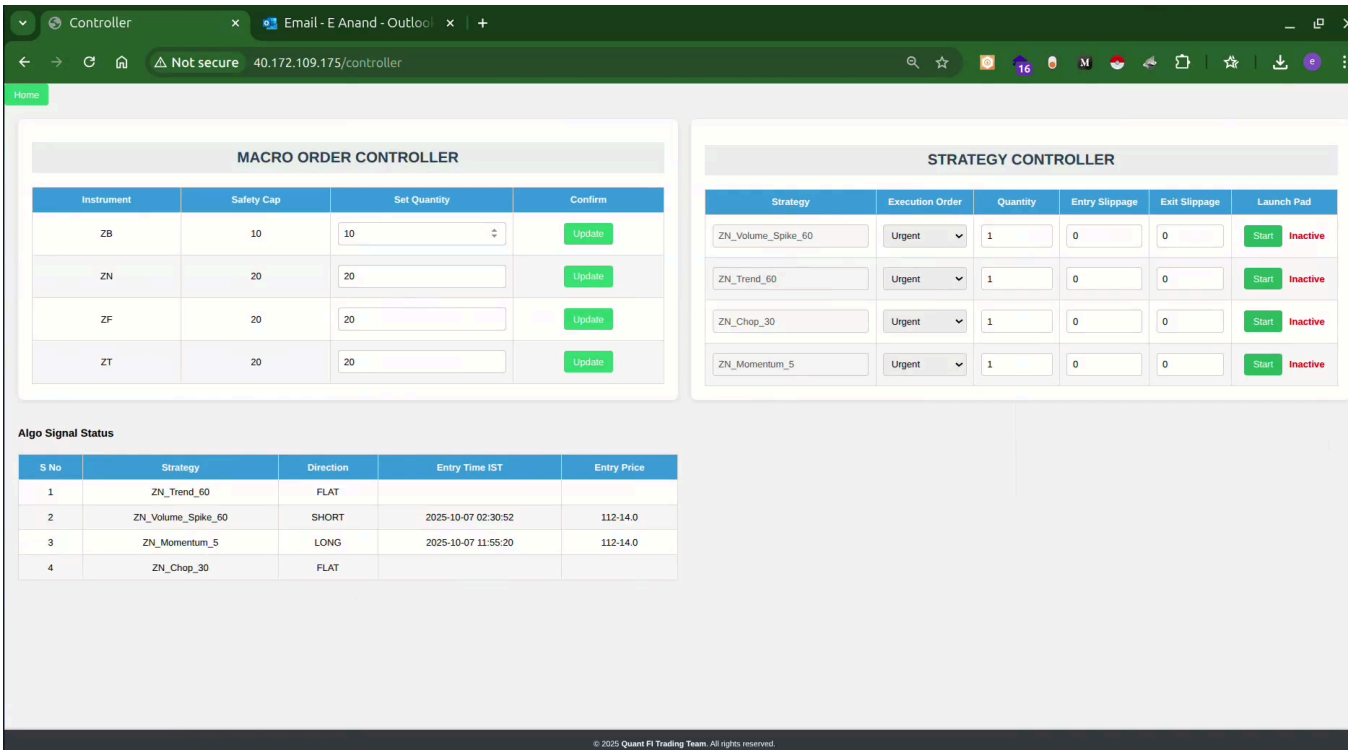

# 10. Algo Controller

## Maximum quantity protection

The Algo Controller provides a configurable maximum quantity limit. This acts as a safety boundary before an order reaches execution, preventing an erroneous quantity value from being submitted without passing the configured cap.

## Algorithm execution controls

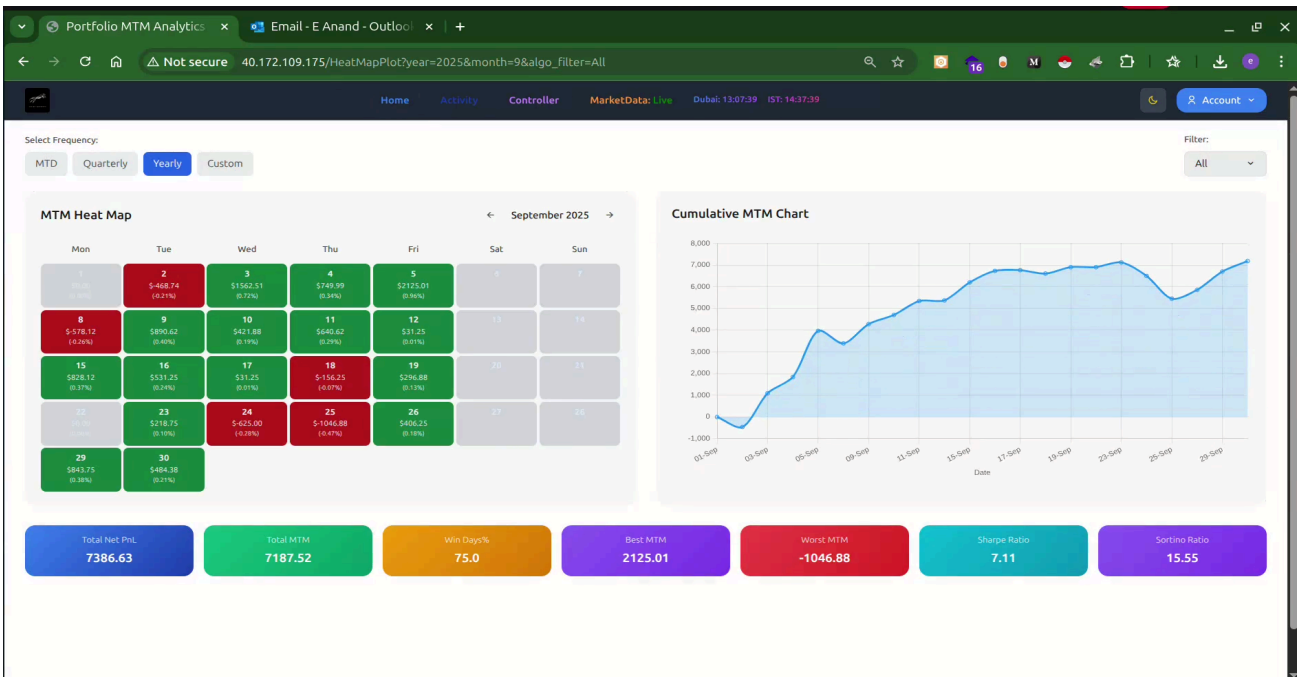
- Algorithm start control.
- Entry slippage configuration.
- Exit slippage configuration.
- Execution parameters intended to improve practical fill quality.




# 11. Portfolio Analytics

- Portfolio heatmap visualization.
- Cumulative MTM analysis.
- Multiple performance metrics.
- Strategy-wise filtering for performance analysis.

The analytics layer allows aggregate portfolio performance to be decomposed by strategy rather than viewed only as a single portfolio-level result.



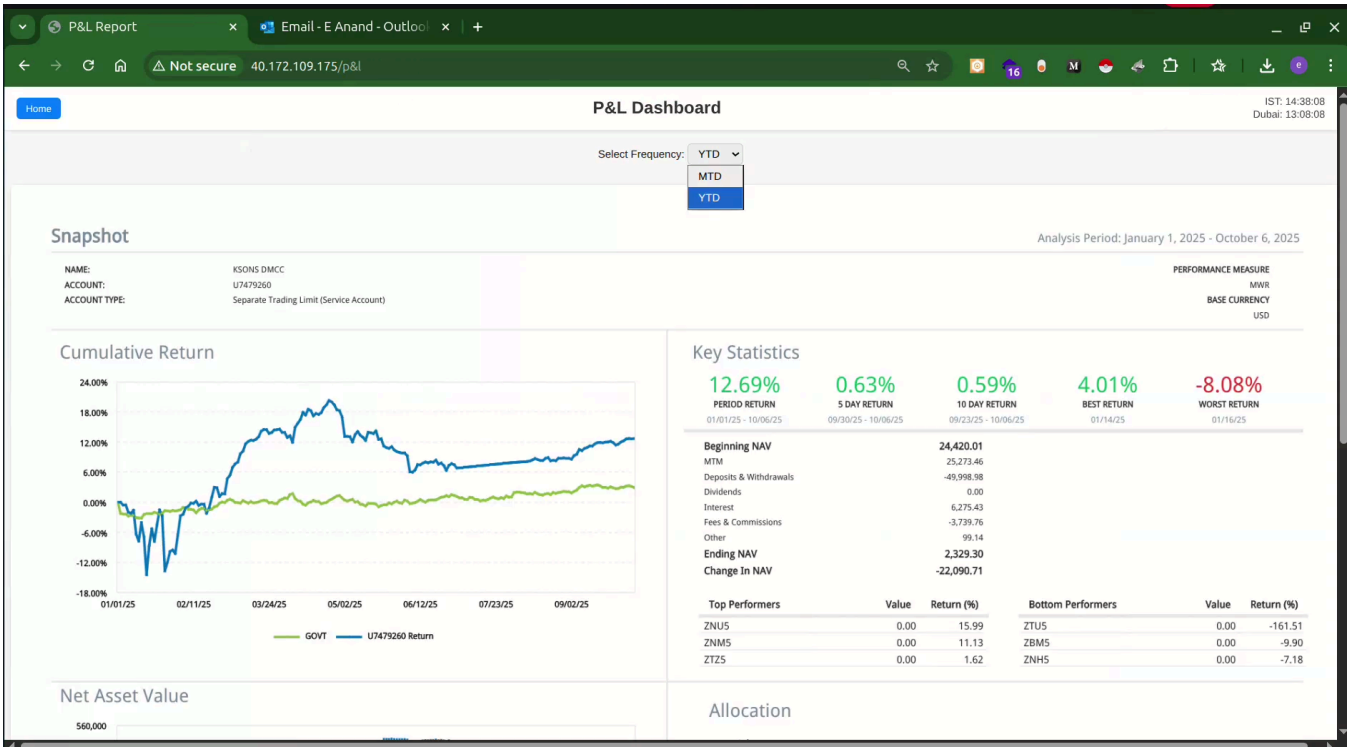
# 12. IBKR Account & P&L

## Account section

- Account value retrieved from IBKR.
- Account-level information presented within the platform.

## P&L dashboard

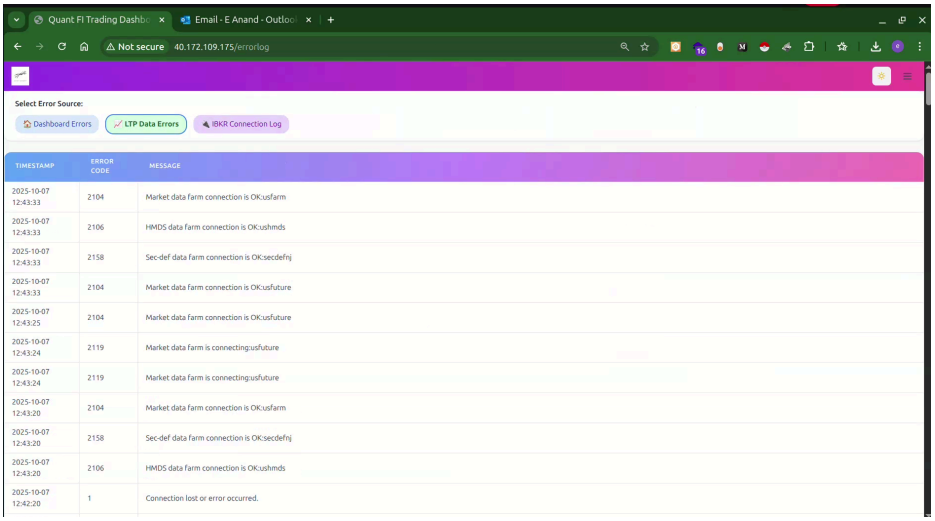
- Daily P&L.
- Realised P&L.
- Unrealised P&L.
- YTD P&L.
- MTD P&L.
- Live broker-sourced P&L visibility.




# 13. Error & Operational Monitoring

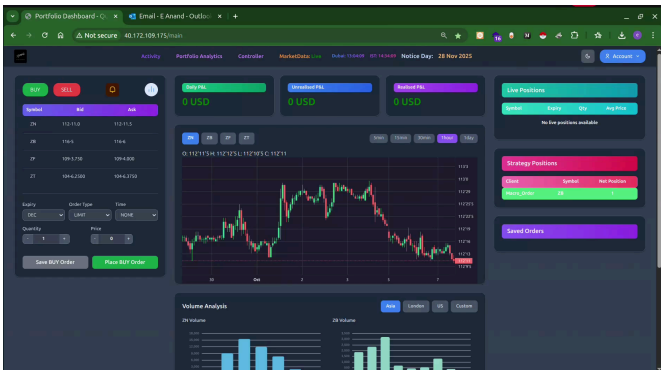
The application provides a dedicated error-log area so operational problems can be inspected from within the platform rather than requiring users to rely exclusively on backend logs.

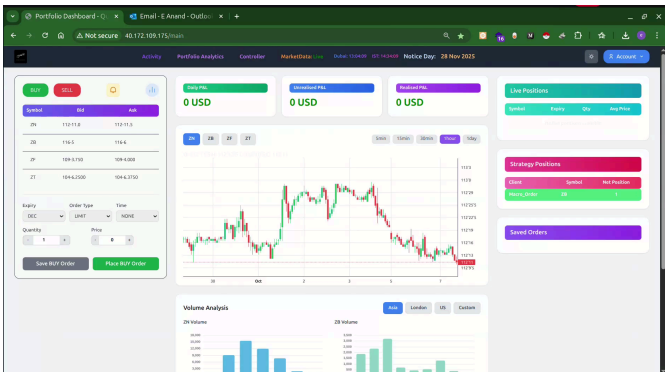
- Dashboard errors.
- LTP/live market-data errors.
- IBKR connection errors.
- Source-based filtering.
- Additional system-level error information where available.




# 14. UI/UX Design

- Light and dark modes.
- Trading-oriented three-column dashboard.
- Market information and execution controls kept within the same workspace.
- Direct access to instruments, bid/ask, Level 2, charts, P&L and positions.
- Navigation organized around Activity, Portfolio Analytics and Algo Controller.



# 15. End-to-End User Workflow

1. Login

2. Open trading dashboard

3. Select instrument

4. Review live bid/ask and Level 2

5. Review day high, day low and previous close

6. Analyze live chart and session volume

7. Prepare or load an order

8. Configure quantity, order type and timing/expiry

9. Save or place order

10. Monitor/edit open order

11. Monitor live and strategy positions

12. Monitor P&L

13. Control algorithm execution

14. Review portfolio analytics

15. Inspect system errors when required

16. Logout

# 16. Engineering Highlights

The project should be evaluated as a trading operations and execution system rather than merely as a Django dashboard. The technically meaningful components are the integration of live broker data, market-depth visualization, order workflows, algorithmic controls, portfolio analytics and operational monitoring.

- Real-time broker/market-data integration.
- Level 2 order-book handling and visualization.
- Manual order lifecycle and open-order management.
- Live position and strategy-position monitoring.
- Execution safety through configurable quantity limits.
- Algorithm execution controls with entry and exit slippage parameters.
- Live P&L and account-value integration.
- Strategy-level portfolio analytics.
- Trading-session volume analysis.
- Automated price-alert notification through SMTP.
- Role-based application structure.
- Operational error and connection monitoring.